

TransMedics Group

NASDAQ: TMDX



Price: \$105.13

TP: \$195.29 (49.25%)

SL: \$101.44 (-22.47%)

Investment Horizon: 2+2 Years



Healthcare

Sector Head: xxxxxx

Analysts: xxxxx, Daniel Bhardwaj, xxxxx, xxxxxx, xxxxx, xxxxx, xxxxx

Company Overview



TransMedic is a medical technology company transforming organ transplant therapy for end-stage organ failure patients across multiple diseases.

Value Proposition

TransMedics offers a fully integrated, end-to-end organ transplant solution.

 TMDX provide **end-to-end solutions for organ transplantation** through its OCS and its National OCS program, which helps streamline operations on the clinical side.

 They operate a network of **22 aircraft and 18 hubs across the US** provide aid on their logistics side helping them to reach patients faster than competitors.

 Their OCS technology maintains donor organs in a near-physiological, warm, beating condition during transportation ensuring **longer preservation times and improved post-transplant outcomes.**

Product Offerings

TMDX's Main Product Offerings.

Heart	Lung	Liver
Second most used service at 854 cases accounting for 18% of the US market.	Currently undergoing rebranding as part of the DENOVO lung trial to capture new revenue.	Main revenue stream, covering 4,197 cases in 2024 accounting for 36% of the US market.

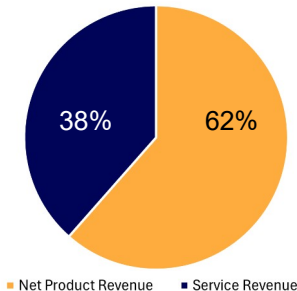
Revenue Breakdown

Revenue Streams.

Net product Revenue: Disposable OCS device sets used per transplant case. TMDX's highest margin revenue at 79% (gross).

Service Revenue: TMDX's fastest growing revenue line, encompassing NOP Clinical Services, and Transport Logistics.

Revenue by Segment (FY2025).



Recent Development/ Future Projects

How TMDX have, and plan, to grow.

-  **DENOVO Lung Trail** - TransMedics received FDA approval for the OCS enhance heart and **DENOVO lung trails**, with both trials initiated in Q4 2025 to drive heart and lung adoption.
-  **OCS Enhance Heart Trial** - FDA-approved trial expanding donor heart time and distance limits. Aims to **demonstrate superiority vs cold static storage.**
-  **OCS Kidney Program** - TMDX is advancing their Kidney program towards and **FDA trial in early 2027**, targeting the **8,000 to 9,000** kidneys currently discarded annually in the U.S.
-  **European Expansion** - TMDX is prioritising the launch of the **NOP model in Europe**, starting with Italy, which has the potential to nearly double the company's TAM.

Management

Stable expert governance anchoring specialisation of management roles.

Founder & CEO

The strategic anchor providing mission-critical stability.



Dr. Hassanein's **27-year tenure as CEO** and **1.8% equity stake** establish a stable, founder-led foundation aligning leadership with **long-term, mission-driven investment** case.

As an **award-winning cardiothoracic surgeon** and primary **inventor of the Organ Care System (OCS)**, Dr Hassanein anchors TransMedics with the unmatched **technical credibility and clinical authority** required to convert global hospital boards and redefine the standard of care.

Board Governance & Veteran Oversight

Guided by S&P 500 veterans to ensure disciplined scaling.



Chaired by Jim Tobin who **grew Boston Scientific's revenue by \$8 billion** through a global medical market expansion. His proven track record provides the **strategic blueprint for TransMedics' current international transition**.



Audit Chair Merilee Raines **brings financial discipline** to the board. As the former CFO of IDEXX Laboratories, she ensures TransMedics replicates the IDEXX model of **high-margin, recurring service revenue** through their OCS Program.



An exceptional **12.6-year average board tenure** provides stability whilst their operational expertise ensures that while the executive layer evolves to manage global expansion, the **long-term strategic integrity remains uncompromised**.

Sources: [Transmedics Leadership](#), [Board Committee](#)

CCO Specialisation: Domestic vs. International

Decoupling U.S. "Industrialisation" from Global Expansion to de-risk growth.



Giovanni Cecere (CCO)

- 20+ year veteran and core patent holder; has technical mastery of OCS Program.
- Spearheaded transition to automated production and integration of aviation fleet.



Dr. Tamer Khayal (SVP of International)

- Former CCO, Architect of US OCS Program and now tasked to replicate success abroad.
- Cairo University-trained MD and 25-year veteran, expert in global regulatory.

Operational Outlook: The "Divide & Conquer" Model

U.S. Operational Efficiency (Cecere).

Fleet Scaling → Maximised Service Margins

Automated Production → Operating Leverage

Domestic De-Risking → Stable Capital for Global Expansion

Global Growth & Replication (Khayal).

Clinical Diplomacy → Regulatory Capture

Navigating Legal Frameworks → Unlocking the European Market

Global Replication → Total Addressable Market Capture

Industry Overview

A rapidly growing and specific demand of which TransMedics is happy to accommodate.

End of UNOS (United Network for Organ Sharing) Monopoly

A Massive Vacuum in the Market forms.



Upon a vote from congress in '23. The UNOS **Monopoly** has officially ended. Moving to a **Multi-Vendor** structure.



UNOS was found to be **untenable**. It had **record high** discard rates for organs, combined with **conflict of interest** in the board of UNOS being the same as the OPTN and their proprietary code in UNet **failed to modernise**.



New efficiency **mandates** plays into TransMedics' OCS tech as well as their NOP Connect system using modern API systems now **required**.

Tailwinds

Headwinds

Key drivers and challenges impacting performance and expansion.



February '26 reform mandates real-time organ tracking. TransMedics has the **most advanced systems**.

The OPTN ETF has set a goal of **60,000** transplants.

ENHANCE Heart and DENOVO Lungs have **received IDE from the FDA**.



Pilot **shortage** and **price hike**. There is projected to be 80,000 unfilled pilot positions by 2032.

Changes to medicare with the IOTA system increases hospitals price **sensitivity**.

Announced **margin compression** due to European NOP launch (beginning in Italy).

Growth and Shift

The market is moving to NMP.

9.6% CAGR
For global
transportation

2x that of typical
Medical devices

53% market
share and 15.9%
CAGR for NMP

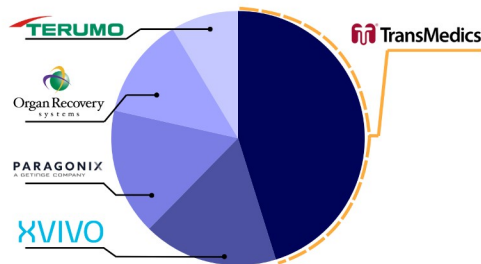
3x the growth rate of
typical storage methods

36% of U.S. Liver
Transplants

Up from 26% in '24;
targeting 50% by '27

Competitor Landscape

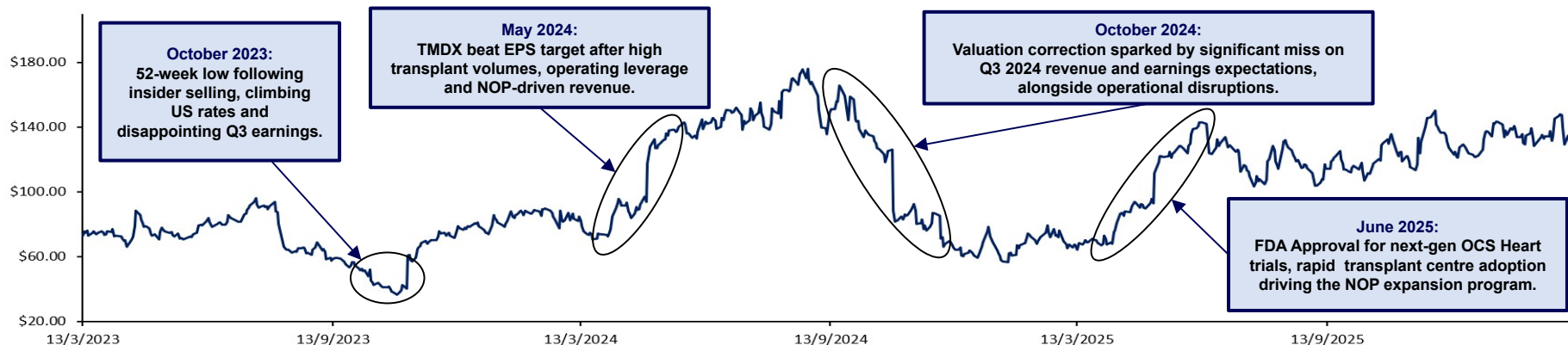
How TMDX fairis to the rest of the Industry.



TransMedics maintains a **near complete monopoly** on NMP systems. That in combination with it's **full-stack service** (Trident) of extraction, maintenance and delivery allows it to maintain a **near 50% revenue** share of the market and **10% of all organ** transplants in the US.

Why the Opportunity Exists

The market underestimates TransMedics' shift to end-to-end coverage as well as its plans to expand into new markets.



Why It May Be Mispriced

Reason I: Transition to Transplant-as-a-Service

TMDX has evolved from a traditional device seller into a “transplant-as-a-service” provider, generating **recurring, per-procedure revenue** rather than one-off sales. However, the market still appears to value it as a capital equipment company.

Reason II: Consistent Earnings Surprises

TMDX has consistently beaten earnings expectations, with **EPS surprises of 40% in Q4 2025 and 78% in Q3 2025**. This suggests the market is repeatedly underestimating its growth.

Reason III: Expansion Opportunities

TMDX have **several strategic growth initiatives** lined up for 2026, to **catalyse growth** short, mid- and long-term. These are not yet reflected in current expectations.

Investment Theses Summary

1

Thesis I: End-to-End Coverage

TMDX have developed a model in which they cover all aspects of **organ care, transport, and transplantation**. This acts as an **end-to-end** service that takes care of patients across the stages of organ transplants.

2

Thesis II: International NOP Expansion

TMDX plans to **expand its proven US transplant service model** into Europe. This creates a significant growth opportunity by increasing transplant volumes, capturing recurring revenue, and tapping into a **large, underpenetrated market**.

3

Thesis III: Enhanced Market Penetration

TMDX plans to **expand on its current product and service areas** to promote growth. This will be achieved through an increase in the level of approval by the FDA, and also through label expansion.

Thesis I: End-to-End Coverage

TransMedics produces a superior product and service.

What is the Industry Standard?

Cold Box.

The cold box method, used for 50 years, flushes the organ with a **cold preservation solution** and it is then placed in a **plastic bag** and submerged in an **ice slurry** inside a **standard commercial cooler**. Relying on **Hypothermia** to reduce oxygen and nutrient needs by 90%.

This **does not stop** the organ from **dying**, simply slowing its death. Due to lack of blood flow the organ begins to suffer from **ischemic injury** **immediately**. Most commonly: **Necrosis**.

Success Rate

Cold Box		NMP
61%	Livers	98%
21%	Hearts	97%
20%	Lungs	96%

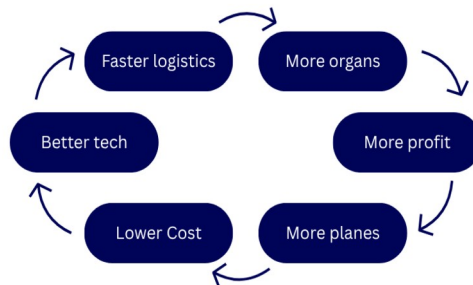
How they Maintain Margins

The Triple Fee Model.

The Kit - A brand new single use perfusion kit is required.

The Surgeon - Transmedics NOP teams handle extraction.

The Airline - what would be third-party profits goes directly to Transmedics.



What is the Product?

Out with the Old in the with New.

Pumping 1.1L of warm donor blood infused with a **proprietary nutrient solution** keeps the heart running.

Clinical Evaluation can take place **during transportation**. Checking **lactate levels** for the heart, physically **inflating a collapsed lung** or monitoring **bile production**.

NOP connect is a mobile and web-based application that allows for **GPS tracking** and **ETA**. Real-time data to the recipient surgeon and handling of paperwork for **HIPAA-compliance**.

The Fleet

I Bought the Airline.



TransMedics acquired Summit Aviation in August '23. Buying both the **jets** and the award-winning **flight school**.



Owning the flight school allows for the **specialised high-intensity training**, specifically **mountain flying training** and allowing for training **tailored** to TransMedics needs.



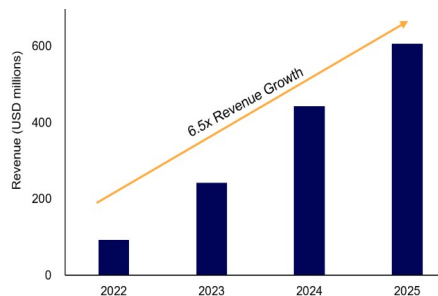
Exclusive flight training partner for Gallatin College, allowing for the scouting of **top talent** and a direct source to **counter** the **scarcity** produced by COVID.

Thesis II: International NOP Expansion

TransMedics plans to expand into new regions, starting with Europe, by replicating its successful US business model.

Current Position in the US

TransMedics has established a very strong and scalable position in the US.



TransMedics is already a **market leader** in organ transplant technology and services in the US, having **completed over 5,139 OCS transplants** in 2025 (+38% YoY).

With this **proven and scalable model**, they are now **expanding into Europe** as their next growth phase. To lead this, is former CCO Tamer Khayal who was recently appointed as **SVP of International**.

Europe - An Untapped Market

Expansion into Europe will increase TransMedics' TAM.



Italy serves as the first step towards a **broader multi-country European network**, acting as a pilot before scaling its NOP model across the rest of Europe. This represents a significant untapped opportunity for TransMedics, as **static cold storage remains the clinical standard in Europe**, accounting for **45% of organ preservation** in 2025.

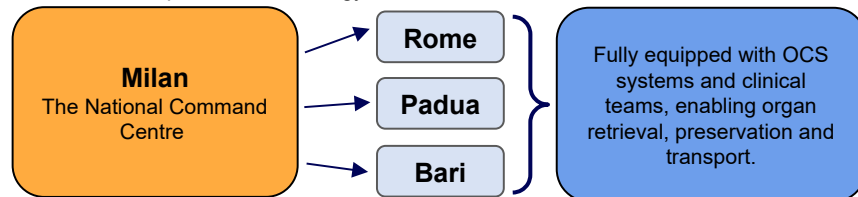


This positions TransMedics as a **first mover in delivering an integrated transplant service model in Europe**, targeting a **market of over 42,576 annual transplants**. TransMedics can capitalise on this by expanding transplant volumes, attract new hospital clients and capture higher recurring revenue.

Italy

Italy is TransMedics' first step in its expansion into Europe.

In September 2025, TransMedics announced plans to **launch its first European NOP system in Italy**, centred around **four regional hubs**. This is to be completed by H1 2026. Italy, specifically Mirandola, is recognized for its **excellence and specialization** in the biomedical and perfusion technology sector.



Mercedes-Benz Partnership

TransMedics' partnership with Mercedes-Benz will drive the rollout of its logistics network.



Mercedes-Benz

In September 2025, TransMedics **partnered with Mercedes-Benz** to launch **Italy's first dedicated organ transport network**. This includes a **fleet of custom V-Class vehicles**, (used as ambulances across Europe) deployed across the 4 NOP hubs to support the rollout of its service model. The network is designed to **enable 24/7 organ transport and coordination**.

Transport is pivotal to TransMedics' model, with **22 aircraft in the US supporting over 5,000 OCS transplants annually**. This partnership will increase transplant capacity, reduce delays, and perform more procedures.

Thesis III: Enhanced Market Penetration

TMDX are aiming to double the number of transplants annually by 2028, capturing a greater market share in heart, lung, and liver transplants.

Heart

How they are expanding their market share in the Heart segment.



TMDX currently capture **18% of US heart transplants** (854 cases in 2025).



ENHANCE Heart Trial - Part A is focused on supporting prolonged heart perfusion using the OCS heart system. This **extends the viable transport window from 4 hours**, allowing for the organ to be taken a greater distance.



ENHANCE Heart Trial - Part B is designed to demonstrate the **superiority of the OCS compared to cold storage**, directly targeting potential growth within this segment.

Lung

How TMDX are breathing life into a neglected segment.

Market Share

Currently **only have a 2%** market share within the US Lung transplant market, only 88 cases in 2025.

Constraint

This is due to **label constraints imposed by the FDA** when the product initially launched in 2018, restricting the OCS from a range of different transplants.

Approval

In Jan 2026, their DENOVO trial came to completion where the **FDA issued unconditional approval for their Lung OCS** making this become a high growth potential segment.

Liver

Deepening their already dominant position.



TMDX currently capture **36% of all liver transplants in the US** making it their most penetrated segment thus far.



Management still claim that they are 'early in the liver adoption market' placing them in a strong position to excel over the coming years.

Label Reinforcement

DCD Expansion

DBD Expansion

Kidney

TMDX plan to introduce another revenue stream.



TMDX are planning further expansion of their **OCS products into Kidneys** with an **expected launch in 2029**. While this is well into the future, this will provide theme access to a significant new revenue stream.



FDA trials are targeted for **early 2027 with commercial launch in 2029**.



The TAM of the Kidney transplant market is significant with over **90,000 people** on the kidney waitlist in the US alone, facing **wait times of 3-5 years**.

With near **30% of donor kidneys being discarded**, TMDX's OCS will ensure that there are more available donor kidneys, helping to cut down on this waitlist both in the US and Europe.

Comparable Companies Analysis

Growth is driven by increasing organ transplant volume and adoption of the OCS system across transplant centres.

\$ in billions



Ticker	Company Name	Market Cap	Enterprise Value (FY0)	Revenue (LTM)	Revenue (NTM)	Revenue Growth (FY1/FY0)	Forward P/E (NTM)	EV/Revenue (FY0)	EV/EBITDA (FY0)	Operating Margin (PoP Diff)
XVIVO	Xvivo Perfusion AB	\$0.74	\$0.62	\$0.09	\$0.12	34.30%	51.45	7.03	41.68	-3.00%
PODD	Insulet Corp	\$15.21	\$20.23	\$2.71	\$3.47	19.30%	31.79	7.47	26.92	2.90%
LNTH	Lantheus Holdings Inc	\$4.91	\$4.62	\$1.54	\$1.51	14.00%	13.82	3	13.45	-8.80%
PEN	Penumbra Inc	\$13.20	\$11.65	\$1.40	\$1.64	13.60%	62.42	8.3	61.31	5.80%
DXCM	Dexcom Inc	\$25.72	\$24.84	\$4.66	\$5.39	12.10%	25.57	5.33	19.91	5.50%
EW	Edwards Lifesciences Corp	\$48.01	\$45.84	\$6.07	\$6.84	9.80%	27.46	7.56	24.59	-0.20%
INSP	Inspire Medical Systems Inc	\$1.48	\$2.33	\$0.91	\$0.99	8.00%	24.32	2.55	18.03	1.10%
Peer Mean		\$15.61	\$15.73	\$2.48	\$2.85	15.90%	33.83	5.89	29.41	0.50%
TMDX	Transmedics Group Inc	\$3.61	\$4.18	\$0.61	\$0.77	18.30%	38.48	6.91	26.7	9.40%

Sources: [Refinitiv](#)

The Griff Investment Fund

Modelling & Valuation

The DCF points to a 49% upside in the base case.

Discounted Cash Flow Model

	Exit EV/EBITDA	EBITDA (FY2030)	Implied Share Price	Implied Upside
Bull	26.00	183.98	233.37	+78.35%
Base	20.00	179.20	195.29	+49.25%
Bear	18.00	111.41	101.44	-22.47%

Sensitivity Analysis

How the WACC and Terminal Growth Rate effect upside/downside:

		Terminal Growth Rate				
		1.00%	1.50%	2.00%	2.50%	3.00%
WACC	8.19%	203.60	208.52	214.24	220.96	228.97
	8.69%	195.19	199.39	204.22	209.83	216.43
	9.19%	187.54	191.17	195.29	200.04	205.54
	9.69%	180.54	183.69	187.25	191.30	195.95
	10.19%	174.09	176.84	179.93	183.42	187.40

Key Model Assumptions

Line Item	Projection (FY2030)	Drivers
Revenue		The bull case assumes transplant centres adopt OCS rapidly , whilst the bear case considers FDA regulation slowing TransMedics' penetration into the kidney market.
COGS		COGS rise across all three cases given European expansion plans. The bear case accounts for increased fuel prices impacting service revenue.
Operating Expenses		Opex are hit by increasing clinical trial costs (kidneys, heart and lung) to drive demand in OCS . Modelled aggressively, they sit on the upper end of historical operating expenses.
CAPEX		Capital expenditure is assumed to increase across all cases as a result of the EU and fleet expansion , maintenance and R&D in kidneys over three years, with the bear case considering poor ROI.

Risk Analysis

As a distinct market outlier, TransMedics balances the complexity of its aviation logistics with a statistically stable and superior performance profile.

Context from Risk Analysis

Potential benefits:



TransMedics' average return being **4.5x the peer mean** highlights a **significant performance gap**. This confirms the thesis that TransMedics is not merely capturing market share but is **fundamentally expanding TAM** through superior market penetration.



Sharpe ratio of 1.05 is **meaningfully higher** than **portfolio average of 0.45**. This **superior risk-adjusted** profile validates the **stability of the OCS program** bringing statistical confidence to the 2026 European expansion.



A remarkably low R^2 of 0.09 reveals that **91% of price movement is independent of broad market trends**. This signals a **logistics moat** where value is driven by internal milestones rather than macro-healthcare shifts.

Potential drawbacks:

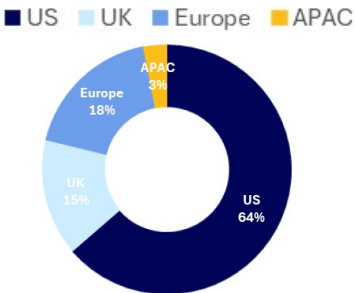


Elevated volatility is a byproduct of the capital-intensive aviation fleet. However, implied risk is offset by a **near-normal kurtosis** and a superior Sharpe ratio, a profile that **compensates investors for volatility**, unlike direct competitor **XVIVO who carries higher volatility and a negative Sharpe ratio**.



While a $\beta > 1$ suggests a sensitivity to broad market sentiment, the 0.09 R^2 implies that while TransMedics does react in-line with the market, most of **price movements are accounted for by its idiosyncratic operational performance**.

Share Recommendations and Portfolio Diversification



Comparable Risk Metrics

Company	Sharpe	Beta	R ²	Vol	Kurt	%AR
XVIVO	-0.01	1.13	0.14	57.53	5.94	2.42
Penumbra	0.59	0.84	0.11	32.12	5.18	22.02
Lantheus Holdings	0.44	0.68	0.04	53.60	4.86	26.42
Mean	0.24	0.85	0.12	44.80	5.34	15.03
Median	0.15	0.98	0.11	42.98	4.86	7.83
TransMedics	1.05	1.24	0.09	56.89	2.91	66.78

Risks, Mitigants, & Catalysts

*TransMedics faces various **miscellaneous** and **idiosyncratic** risks all of which it plans for and **mitigates** very well.*

Risks

1

Execution Risk in Europe

What works in the US doesn't guarantee success in Europe. There is **multi-country** regulation and logistical complexity that could prove difficult when expanding. Timeline slips and revenue delays may cause valuation compression.

2

Fuel Volatility

Fuel price volatility is a key risk for TransMedics given its reliance on a fleet of **20+** aircraft. The company has already experienced a **140%** increase in fuel costs, highlighting how exposed its cost base is to fluctuations in energy markets.

3

Regulatory Risk

Expansion depends on navigating complex **regulatory** pathways and securing reimbursement approvals across multiple regions. Without consistent funding mechanisms, hospitals may **resist** adopting a higher-cost solution.

4

Huge Capital Intensity

The business model requires **significant** investment in logistics, aircraft coordination, and specialised clinical teams. If utilisation rates are low, fixed costs can quickly **erode margins** as the company scales.

Mitigants

1

Anchored in Italy

Instead of spreading themselves thin across Europe, TransMedics **anchors** itself in Italy creating a **hub** and **spoke** model around it. This creates operational density so they can stress-test before replicating elsewhere.

2

Route Optimisation and Hedging

The overall cost of fuel remains relatively **small** compared to the total value of a transplant procedure, limiting sensitivity. As the business scales, improved route optimisation and hedging strategies will offset cost pressures over time.

3

Clear Trajectory

Align **regulatory** and **reimbursement** strategies early, rather than sequentially. Position OCS as a cost-offsetting solution by reducing complications, shortening hospital stays and increasing transplant throughput.

4

Emphasise ROI

Leverage **partnerships** to reduce capital intensity and centralise operations through command hubs. Focus on maximising utilisation per system and per transport asset to **drive** efficiency and operating **leverage** over time.

Catalysts

2026- 2027
Italy Launch
and Initial EU
Proof Point

2027-2028
Broader EU
Expansion

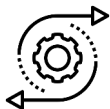
2027-2029
Fleet
Utilisation
Inflection

2028-2030
Kidney
Program
Rollout



Q&A

Appendix - How TransMedics source revenue



Transmedics operates on a hybrid model - a program level partnership with transplant centres, but individual cases are billed on a per-use basis. There are no long term exclusive contracts in this traditional sense, but the relationship behaves like one.



Their commercial team targets the 250+ certified transplant centres across the US, those that are legally approved by UNOS to perform transplants. The sales process involved clinical education, surgeon engagement, and institutional approval. This allows TMDX to convince them that their NOP model is superior to their existing in house procurement approach.



Once a transplant centre is enrolled in the NOP, revenue is generated on a **per-mission basis**. Every time a surgeon at that centre calls on TransMedics to procure and deliver an organ, TransMedics earns both product revenue (the OCS disposable set) and service revenue (the clinical team, logistics, and aviation). There is no minimum volume commitment or annual subscription fee publicly disclosed, meaning the centre is not locked in contractually, but is locked in operationally.



Workflow dependency - If a transplant centre were to stop using transmedics as their provider and courier, this would take them years to rebuild this internal and geographical capability. Surgeons, perfusionists, and coordinators would need to be rehired and retrained.



TMDX employ their own surgeons and clinical teams to carry out these transplants. This can consist of Surgeons, Perfusionists, and clinical coordinators.

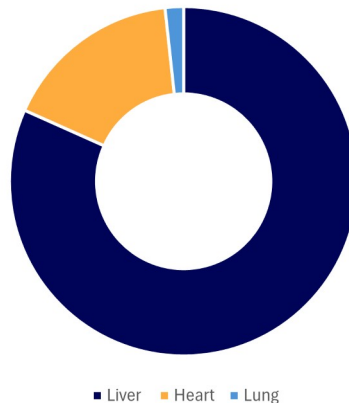
Appendix - Revenue by Organ

OCS case volumes are the primary driver of both product and service revenue streams.

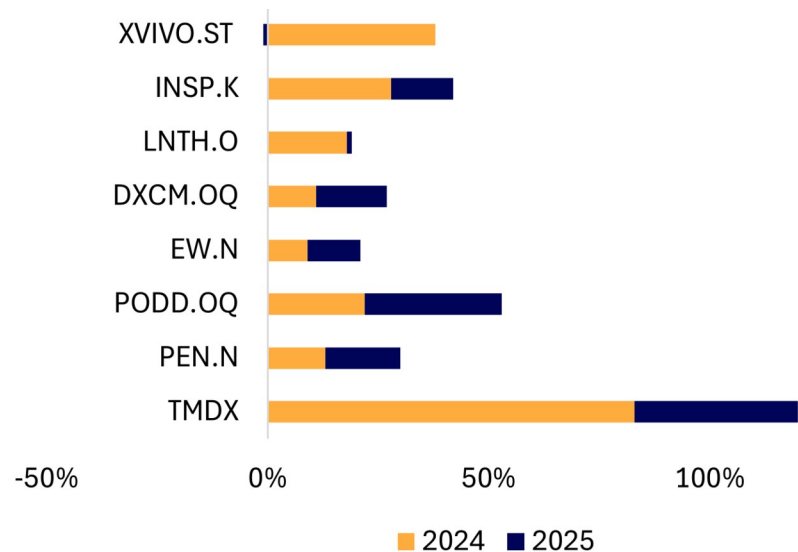
In 2025 - TMDX completed **5,139 U.S. OCS cases** in total - up approximately 38% YoY from 3,715 cases in 2024. Within this, Liver accounted for 4,197 cases (36% of total Liver translate volume), Heart for 854 cases (18% of US heart volume), and Lung for 88 Cases (2% of US lung volume).

TMDX will introduce their Kidney OCS services:

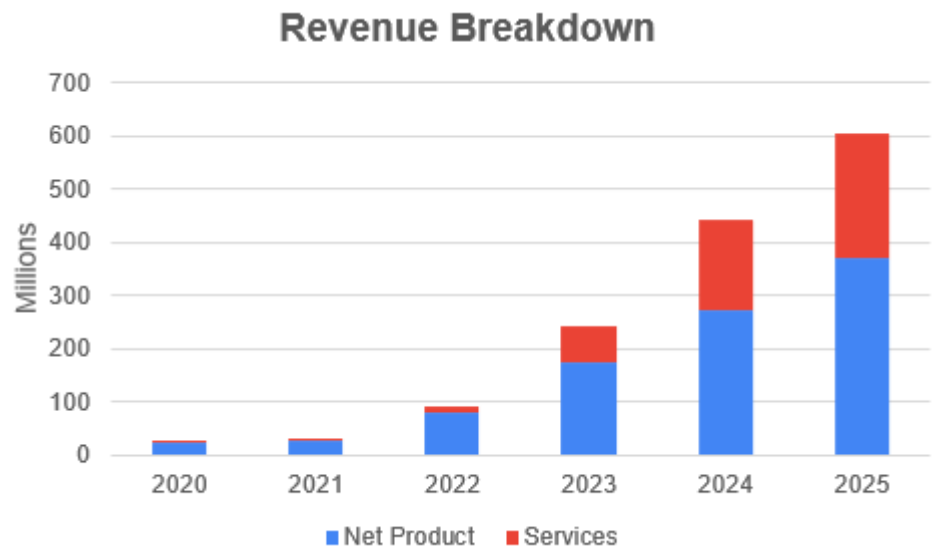
- Goal - to be ready for FDA trial for Kidney by early 2027. Will introduce their next generation OCS 3.0 platform.
- Commercial Launch gy 2029.
- Each year 8000-9000 kidneys are discarded in the US along, their OCS kidney would directly address this.



Appendix - Revenue Statistics

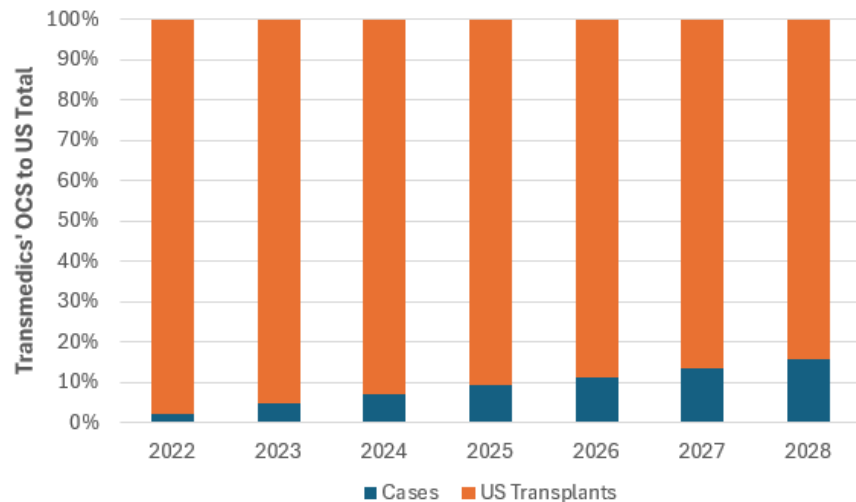


Revenue by Comps
Revenue growth for comparables (2024 - 2025)



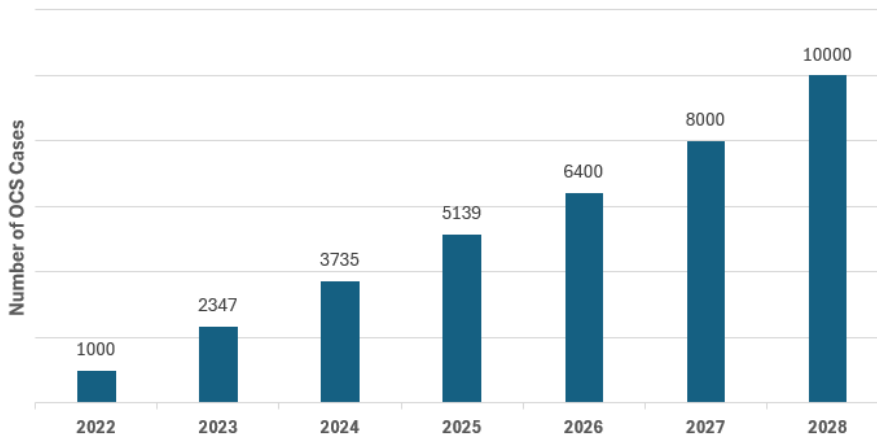
Transmedics Revenue Breakdown by Year
Revenue breakdown by services and products - notice how service revenue has risen exponentially compared to product revenue. This is modelled from in the DCF and based of financial statements.

Appendix - Data on OCS Cases



Increasing Market Share

The share of historical and projected organ transplants Transmedics undertake when compared to the US total across each year.



OCS Transplant Cases

Historical and projected number of OCS cases Transmedics aim to/ have achieved, based on data from 10-K's and forward-looking statements.

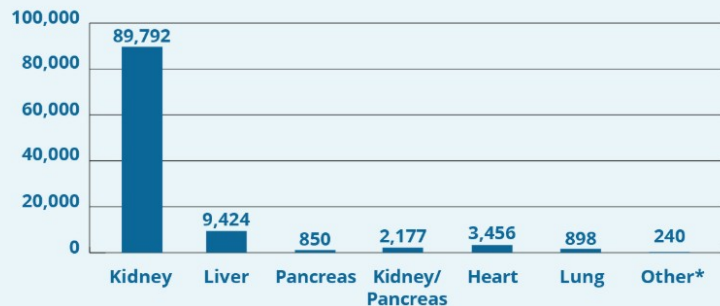
Summary

Both highlight the solid foundation of which Transmedics is ready to be built upon. Cases have risen consistently since specific reporting began in 2022 and they are steadily growing market share.

Appendix - Transmedics' Market

Patients on the Waiting List by Organ

As of September 2024

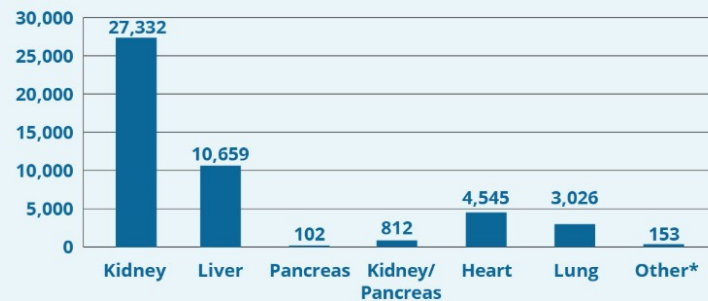


*Other includes allograft transplants like face, hands, and abdominal wall.

Based on OPTN data as of September 15, 2024. Data subject to change based on future data submission or correction. Totals may be less than the sums due to patients included in multiple categories.

Transplants Performed by Organ

In 2023



*Other includes allograft transplants like face, hands, and abdominal wall.

Based on OPTN data as of September 15, 2024. Data subject to change based on future data submission or correction. Totals may be less than the sums due to patients included in multiple categories.

Transmedics' Unique Positioning

Simply, demand outstrips supply across all organs. There are various factors playing into this from location of transplant and organ retrieval, biological viability, manner of death of donors and its impact on the organ itself (DCD - circulatory, DBD - brain), etc.

Transmedics can help solve a lot of these constraints based on logistical improvements, OCS live monitoring, speed of transplant, etc. increasing the probability that transplants can occur and reducing the number of organs that would otherwise be disposed of - directly driving revenue.

(Retrieved from US Government Data)

Appendix - Porter's five forces

Force 1 - Threat of New Entrants - Very Low

- The barriers to entry in the Warm Normothermic organ perfusion are among the highest of any medical technology segment.
- High regulatory barrier - Any new entrant wanting to compete directly with OCS in the SU, must run a full prospective randomised trial, approved by the FDA. This process usually takes 5-8 years and millions in capital.
- Patent barrier - Transmedics 451 patent portfolio extending to 2040s covers device architecture, perfusion solutions, and methods of use, meaning that new entrants cannot simply replicate their technology.
- The only real way that TMDX faces any threats on this from this through a well capitalised takeover by firms such as J&J or Abbott who are looking to expand into this sector.

Force 2 - Threat of Substitutes - Low to Moderate

- The primary substitute to their warm perfusion tech, is the Cold Static Storage (CCS). This is the tried and tested method that has been used for decades. It currently occupies 95% of all organ transplants globally today. TMDX is a first mover in their space hence why their technology has not picked up higher levels of adoption so far.

Force 3 - Bargaining Power of Supplier - Low

- They have flagged performance of third part suppliers and manufacturers as a risk factor in their SEC filings, but this is a standard medtech risk rather than a structural threat.
- Transmedics controls much of their supply chain and only really rely on third party manufacturers for raw materials for their technology so this isn't seen as a major threat at the moment.

Force 4 - Bargaining power of Buyers - Low to Moderate

- There is no functionally equivalent alternative to OCS for warm normothermic perfusion in the US. Transplant centres that want the clinical benefits of warm perfusion have only one FDA approved option
- Once embedded into the NOP workflow, switching costs are extremely high - surgical teams, logistics and clinical coordination would all need to be rebuilt internally.
- TransMedics is reliant on the limited number of customers of a significant portion of revenue, a small number of high volume transplant centres likely represent a disproportionate share of

Force 5 - Competitive Rivalry - Low to Moderate

- There is an extremely low threat of the same technology being used by different companies. Their main competitors of XVIVO, OrganOx and the other aforementioned competitors all operate in the same space but not the same product. This means threat comes from cold storage which is an existing method that is used to keep organs in a usable condition. However, there are essentially no direct threats in regards to others using the same technology.

Appendix - Competitor Landscape

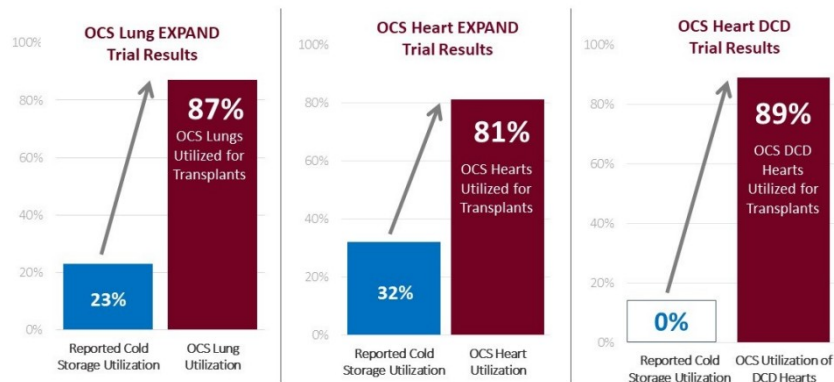
Company	What They Do	Why It Matters to the Table	Why They're Weaker vs TransMedics
Penumbra Inc	Thrombectomy devices (removes blood clots in stroke/vascular cases)	Strong in neurovascular intervention	Mature category, incremental innovation. Doesn't expand procedure volume like organ preservation does
Insulet Corporation	Insulin pumps (Omnipod system)	Large diabetes market, recurring revenue	Competes in crowded chronic care market. Growth is tied to patient switching, not new market creation.
Edwards Lifesciences	Heart valves, critical care monitoring	Market leader in structural heart	Already well penetrated market. Growth is slowing compared to earlier years
DexCom	Continuous glucose monitoring (CGM)	High growth, strong adoption	Still competitive (Abbott, Medtronic) there is a pricing pressure risk and again, doesn't unlock new procedures
Lantheus Holdings	Imaging agents (cardiology, oncology)	Enables diagnosis, not treatment	Indirect value chain exposure. dependent on procedure volume, not driving it like TMDX does
Inspire Medical Systems	Implantable sleep apnea treatment	Niche but growing	Single indication business with very limited scalability compared to a multi organ platform
XVIVO	Organ preservation & perfusion (lungs, kidneys)	Closest direct competitor in transplant technology	Product-focused not service, limited logistics integration, narrower scope vs TransMedics full platform that addresses the whole issue

Appendix - Choosing TMDX over XVIVO

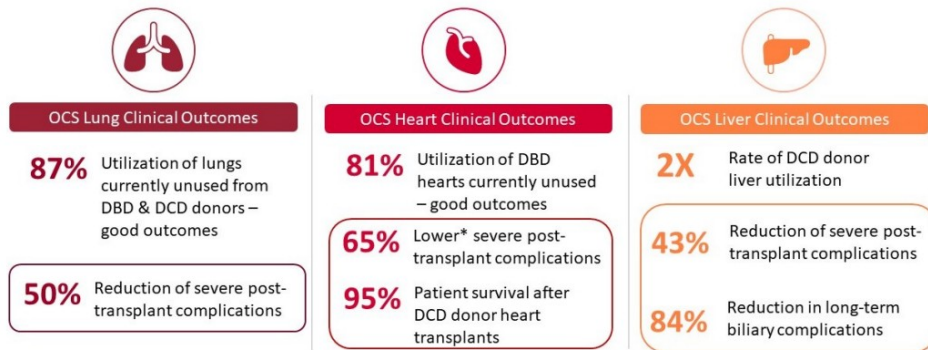
- TMDX are 8x larger by revenue than XVIVO .
 - TMDX are a US domestic company and are breaking in to europe, this means that that are able to receive priority in terms of FDA approval compared to XVIVO who are breaking into the US from europe and so do not receive the same level of treatment. This ensures tht TMDX are able to move at a faster pace in terms of development.
 - XVIVO's XPS lung device carries only a Humanitarian Device Exemption in the U.S. — a lower regulatory bar explicitly stating that "effectiveness of this device for this use has not been demonstrated." [_sec](#) This is a materially weaker commercial position than TransMedics' full PMA approvals for all three organs.
 - XVIVO plans to submit its heart regulatory file to the FDA by Q2 2026, with a full-scale U.S. product launch of Heart Assist Transport and Kidney Assist Transport targeted for 2027. [Daily Political](#) This is still a year or more away from commercial reality in the world's largest transplant market.
 - XVIVO has received FDA Breakthrough Device Designation for its Liver Assist device [Sixsigmaresearch](#), which expedites review — but a U.S. approval and commercial launch for liver is likely 2028 at the earliest.
 - TMDX's current price-to-sales ratio is approximately 7.4x, well below its historical median of 16.8x, suggesting the stock is at a historically cheap entry point relative to its own trading history. [Yahoo Finance](#)
 - XVIVO trades at approximately 57x forward P/E, and its market cap had fallen roughly 48% over the prior year as of early 2026, reflecting significant earnings volatility and uncertainty around the U.S. commercial launch timing
-

Appendix - OCS Utilisation

The OCS Technology Impact on Donor Organ Utilization



OCS Impact on Post-Transplant Clinical Outcomes



Organ Care System

Utilisation stats demonstrate OCS' advantages to standard CCS. Note of 250 transplant centres in the US, about 70 have currently adopted OCS.

(Retrieved from Investor Relations. For more on OCS see <https://www.transmedics.com/ocs-hcp/>)

Appendix - European Expansion

Transmedics already works/have worked with the likes of:



- Royal Papworth Hospital (UK)
- Hannover Medical School (Germany)
- Padua University Hospital (Italy)

UK

Royal Papworth Hospital is a world leader in DCD heart transplantation - they performed a world-first DCD heart transplant in 2019 using OCS, which helped prove preservation, viability assessment outside the body and fundamentally increases the organ pool. This is a strong demonstration of clinical collaboration and technology adoption.

Germany

A major partner in the development and clinical application of TMDX's OCS, Hannover performed studies testing OCS against standard CCS. They demonstrated OCS' functioning states and found them far superior in transport.

Italy

Padua is officially listed as one of TransMedics' international transplant centers using OCS platform for preservation and transplantation. They are one of Italy's major transplant hubs, and are of strategic importance as they bring volume, influence and research, and help tap into the Italian organ market to build a NOP here in the future.

Appendix - A Note on Patents

As of Dec 31, 2025, TransMedics owned and operated 451 issued patents and pending patent applications worldwide, covering the US, Australia, Europe, Canada, China, Israel, New Zealand, and Japan.

The majority of their patents will not expire until 2040, however those that were issued in the late 90s will soon be coming up for renewal or expiry. However, the risk of their new technology being stolen is low due to their Gen 3.0 technology coming out and so these initial patents are no longer in use anyway.